```
import pandas as pd
df = pd.read_csv("Stuudent.csv", sep=";")
```

```
print("=== First 5 rows ===")
display(df.head())
```

```
=== First 5 rows ===
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score
0	female	group B	bachelor's degree	standard	none	72.0	72.0
1	female	group C	some college	standard	completed	69.0	90.0
2	female	group B	master's degree	standard	none	90.0	95.0
3	male	group A	associate's degree	free/reduced	none	47.0	57.0
4	male	group C	some college	standard	none	76.0	78.0

```
print("\n=== Columns and data types ===")
df.info()
```

```
=== Columns and data types ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1002 entries, 0 to 1001
Data columns (total 8 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   gender                       1001 non-null   object
 1   race/ethnicity               1002 non-null   object
 2   parental level of education  1001 non-null   object
 3   lunch                        1001 non-null   object
 4   test preparation course      1002 non-null   object
 5   math score                   1000 non-null   float64
 6   reading score                1000 non-null   float64
 7   writing score                999 non-null    float64
dtypes: float64(3), object(5)
memory usage: 62.8+ KB
```

```
print("\n=== Descriptive statistics ===")
display(df.describe())
```

```
=== Descriptive statistics ===
```

	math score	reading score	writing score
count	1000.000000	1000.000000	999.000000
mean	66.131000	69.220000	68.162162
std	15.152869	14.560275	15.379688
min	0.000000	17.000000	10.000000
25%	57.000000	59.000000	58.000000
50%	66.000000	70.000000	69.000000
75%	77.000000	79.000000	79.000000
max	100.000000	100.000000	145.000000

```
print("\n=== Column names ===")
print(df.columns)
```

```
=== Column names ===
Index(['gender', 'race/ethnicity', 'parental level of education', 'lunch',
       'test preparation course', 'math score', 'reading score',
       'writing score'],
      dtype='object')
```

```
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 1002
Number of columns: 8
```

```
# ----------------------------
# Step 2: Check Missing Values
# ----------------------------
print("=== Missing Values per Column ===")
print(df.isnull().sum())
```

```
=== Missing Values per Column ===
gender                         1
race/ethnicity                 0
parental level of education    1
lunch                          1
test preparation course        0
math score                     2
reading score                  2
writing score                  3
dtype: int64
```

```
# Handle Missing Values without warnings
numeric_cols = ['math score','reading score','writing score']
categorical_cols = ['gender','parental level of education','lunch']
```

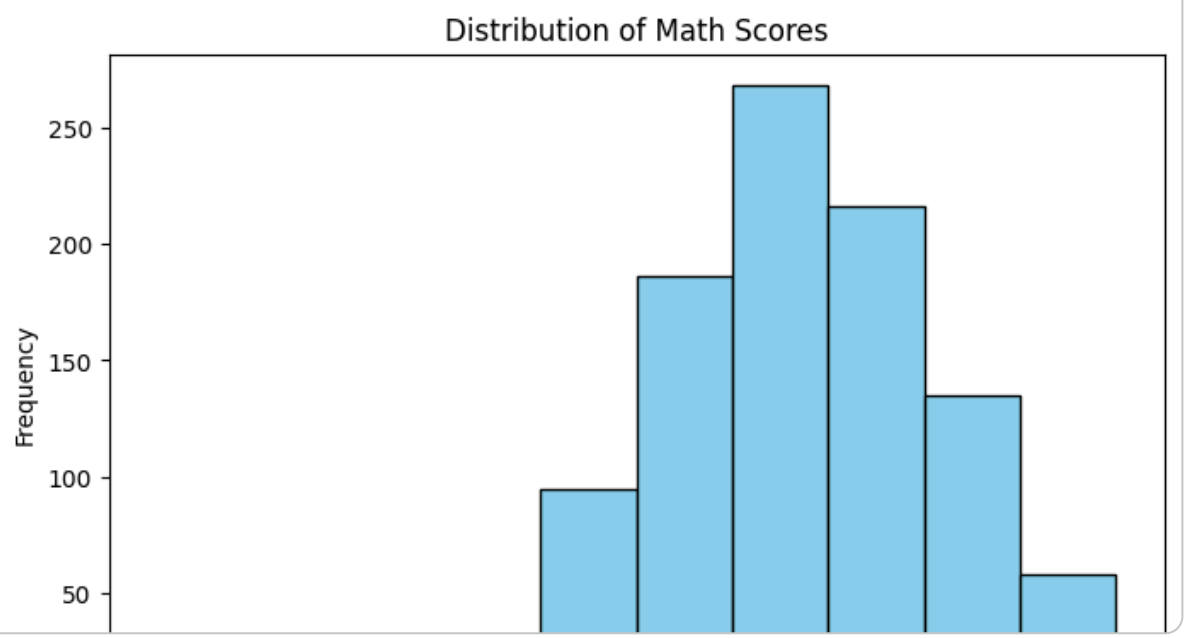
```
for col in numeric_cols:
    df[col] = df[col].fillna(df[col].mean())

for col in categorical_cols:
    df[col] = df[col].fillna('Unknown')

print("=== Missing Values After Handling ===")
print(df.isnull().sum())
```

```
=== Missing Values After Handling ===
gender                         0
race/ethnicity                 0
parental level of education    0
lunch                          0
test preparation course        0
math score                     0
reading score                  0
writing score                  0
dtype: int64
```

```
# ----------------------------
# Step 3: Remove Duplicates
# ----------------------------
print("Duplicate rows before removal:", df.duplicated().
df = df.drop_duplicates()
print("Duplicate rows after removal:", df.duplicated().s
```

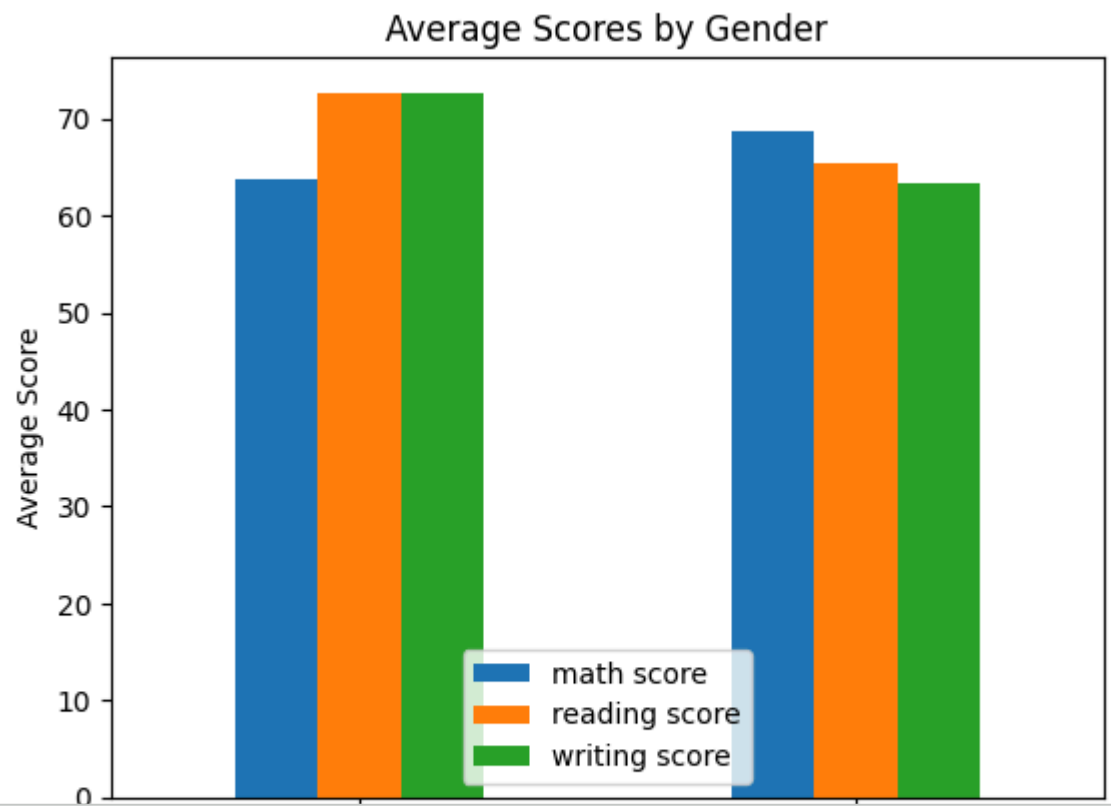
```
Duplicate rows before removal: 2
Duplicate rows after removal: 0
```

```
plt.figure(figsize=(8,5))
plt.hist(df["math score"], bins=10, color='skyblue', edg
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")

plt.savefig("hist_math.png")
plt.show()
```



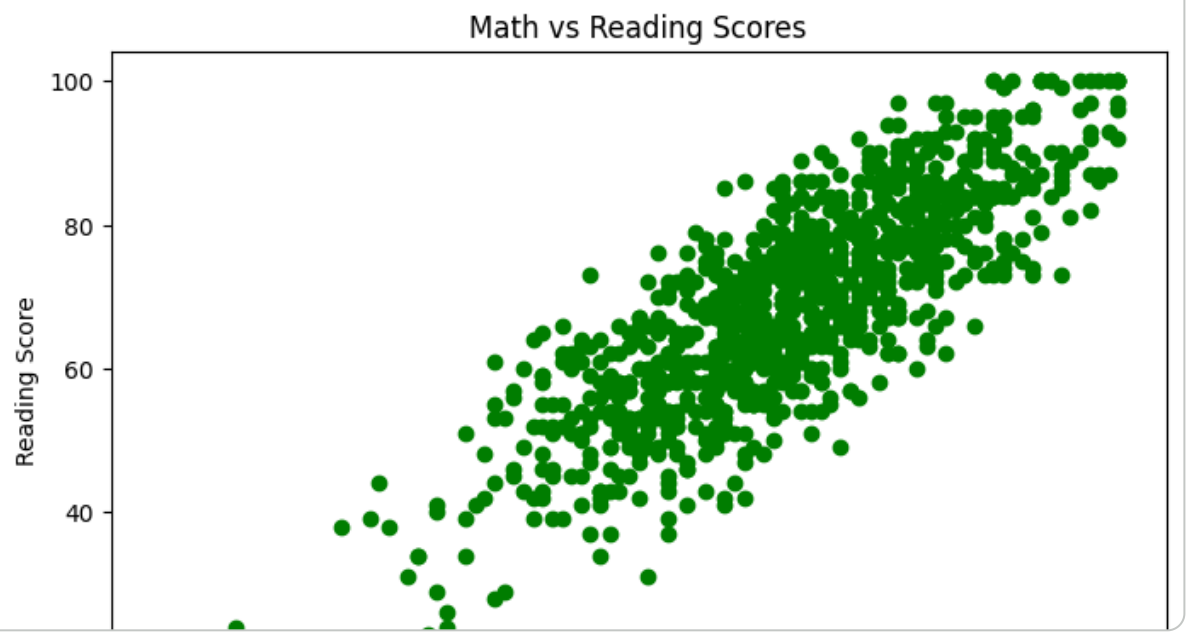

```
plt.figure(figsize=(8,5))
df.groupby("gender")[["math score","reading score","writing score"]].mean().
plt.title("Average Scores by Gender")
plt.ylabel("Average Score")

plt.savefig("bar_gender.png")
plt.show()
```

```
<Figure size 800x500 with 0 Axes>
```

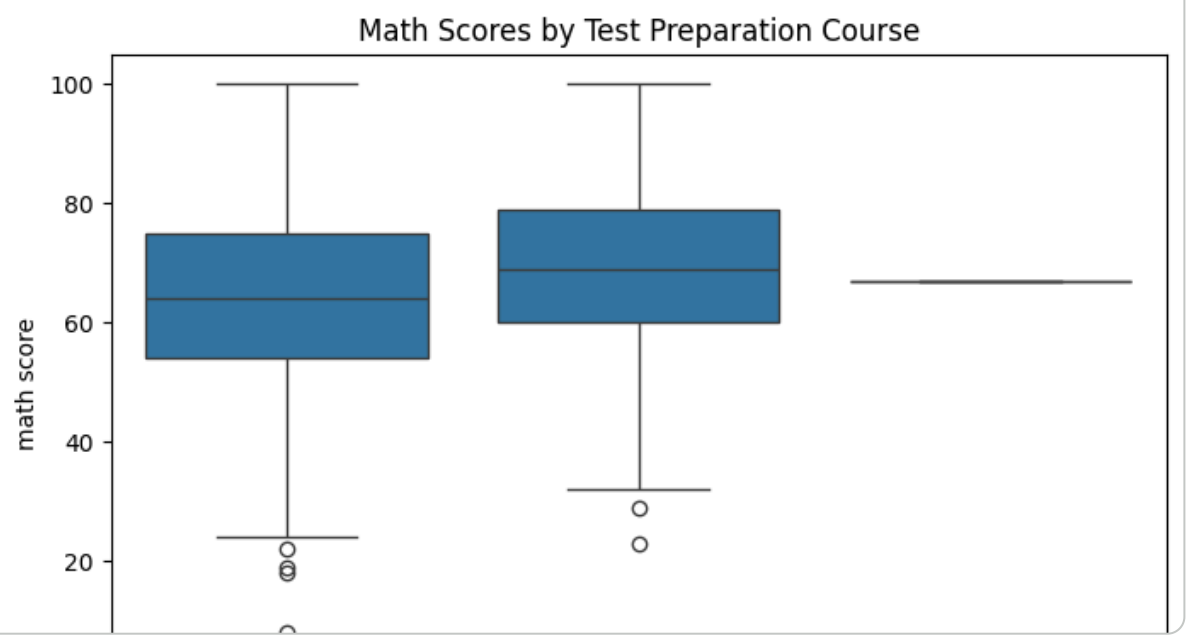



```
plt.figure(figsize=(8,5))
plt.scatter(df["math score"], df["reading score"], color='green')
plt.xlabel("Math Score")
plt.ylabel("Reading Score")
plt.title("Math vs Reading Scores")
```

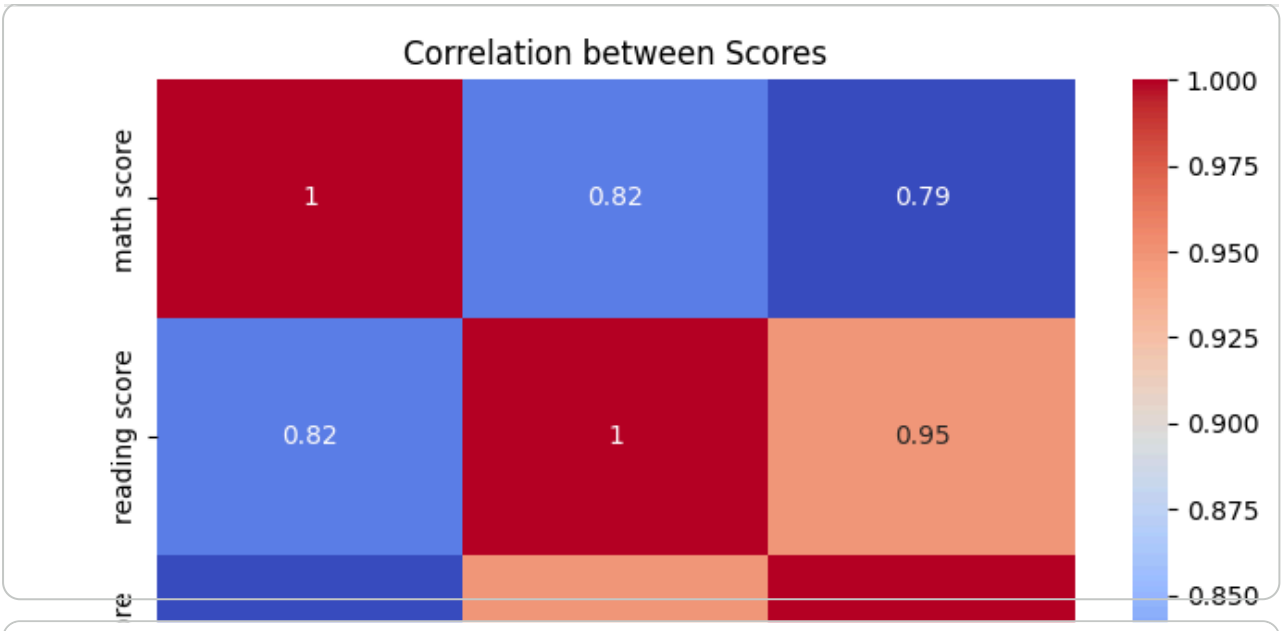
```
plt.savefig("scatter_math_reading.png")
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.boxplot(x="test preparation course", y="math score", data=df)
plt.title("Math Scores by Test Preparation Course")

plt.savefig("boxplot_testprep.png")
plt.show()
```




```
plt.figure(figsize=(8,5))
sns.heatmap(df[["math score","reading score","writing score"]].corr(), annot
plt.title("Correlation between Scores")

plt.savefig("heatmap_corr.png")
plt.show()
```




```
print("""
Patterns and Insights Summary:

Most students scored between 60 and 80 in all subjects.
There is a strong positive correlation between math, rea
Female students performed better in reading and writing,
Students who completed the test preparation course achie
Overall, academic performance across subjects is strongl
""")
```

```
Patterns and Insights Summary:

Most students scored between 60 and 80 in all subjects.
There is a strong positive correlation between math, reading, and writing sco
Female students performed better in reading and writing, while males performe
Students who completed the test preparation course achieved higher scores on
Overall, academic performance across subjects is strongly related.
```